RULE-BASED REGULATORY COMPLIANCE SCREENING FOR PRE-CONSTRUCTION DECISION SUPPORT: CODIFYING THE UDA PLANNING AND DEVELOPMENT REGULATIONS

INDIVIDUAL THESIS

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ABSTRACT

Before any building goes up in Sri Lanka, somebody has to establish what the land will allow. The rules are public: the Urban Development Authority Planning and Development Regulations 2021 set out the permissible floor area ratio, the site coverage, the dimensional minima, the clearances and the approval path. They are also unreadable for the person who needs them, because the binding figure for a plot sits at the intersection of three banded indices across several schedules. Owners therefore find out their planning position after submission, and each returned application costs weeks.

This thesis presents the Regulatory Compliance Advisory engine, Component 3 of the ConstructAI integrated pre-construction platform. A working subset of the regulations is encoded as a declarative rule store of twenty clause-cited rules in four families, together with the full 144-cell floor area ratio schedule. A forward-chaining evaluator closes a fact base assembled from the other components and reports each rule as compliant, violated or not assessed, so a rule whose inputs are incomplete never reports a pass. Verdicts become a buildable envelope and an approval roadmap, and they constrain the floor area the cost component may price.

Over a 1 000-case corpus built to exercise every rule, three findings run against expectation. The rules that bind are not the ones the regulations lead with: rainwater storage, side space, parking and room dimensions all fire more often than floor area ratio, which sits ninth at 14.0 percent. Clearance, not density, is what makes small plots undevelopable, leaving 25.1 percent of a minimum plot buildable at two storeys and 5.1 percent at three. And compliance removes 22.4 percent of drawn floor area on average while leaving 39.3 percent of designs untouched. The corpus is constructed, so these figures characterise the encoded regulations and not the behaviour of real submissions.

Keywords: regulatory compliance checking, planning regulations, rule-based systems, forward chaining, construction decision support, Sri Lanka

ACKNOWLEDGEMENTS

I thank my supervisor for guidance throughout this project, and in particular for one intervention that changed it. Early on I presented the component as a checker that returns pass or fail, and was asked what it would say about a rule it could not evaluate. I did not have an answer. The three-valued design and most of what is defensible about Chapter 6 came out of that question.

I thank my co-supervisor for pressing me on the difference between advice and determination, which changed the wording of every output the component produces.

My three group members made this component possible by agreeing the shared project state before any of us had anything to put in it. The Component 1 owner supplied the plot geometry and the terrain figures, the Component 2 owner the design geometry, and the Component 4 owner consumed what my component produced and told me, more than once, when it was not usable. I am grateful to the practitioners who gave their time to the acceptance sessions and to the Department of Information Technology for the facilities that supported this work.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
BIM	Building Information Modelling
FAR	Floor Area Ratio
FR	Functional Requirement
GFA	Gross Floor Area
IFC	Industry Foundation Classes
JSON	JavaScript Object Notation
NBRO	National Building Research Organisation
NFR	Non-Functional Requirement
NLP	Natural Language Processing
OCR	Optical Character Recognition
PPC	Preliminary Planning Clearance
RQ	Research Question
UDA	Urban Development Authority

1 INTRODUCTION

1.1 Overall project context

ConstructAI began with a complaint we kept hearing while we were still writing the proposal. Four analyses happen before a building is built, four different people do them, and none of the four ever sees the others' work. A surveyor assesses the land. An architect draws the building. Somebody clears the title and the planning position. A quantity surveyor measures the drawings and prices them. Each hands over a document, and a document does not recompute itself when something upstream moves.

What the industry pays for that is overruns. Across a large international sample of projects roughly nine in ten exceeded their approved budget, with mean overruns near 28 percent, a pattern too steady across decades and countries to be written off as estimating noise [1]. Small residential work, which is most of what gets built in Sri Lanka, is not exempt.

Our group built one platform that runs all four analyses over a single shared project record, so the output of each constrains the others. Component 1 turns a land coordinate into an environmental profile. Component 2 turns a floor plan into measured quantities and structural parameters. Component 3, which is mine, checks the resulting design against the planning regulations. Component 4 converts everything into a whole-life cost. Figure 1.1 shows how they sit together.

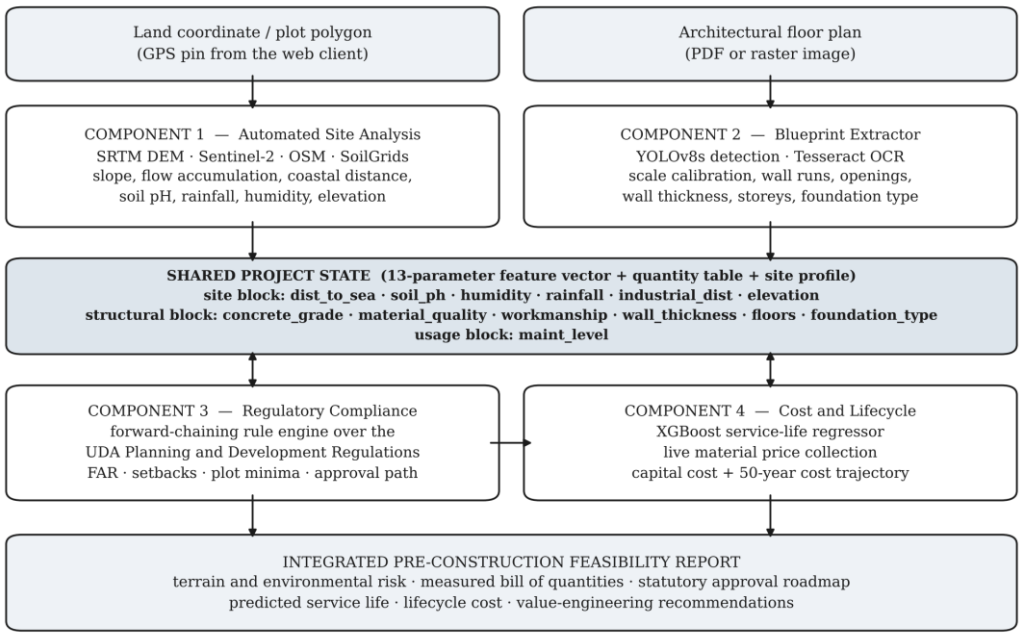


Figure 1.1: Architecture of the ConstructAI platform and the place of Component 3 within it.

1.2 Individual component context

My component is the Regulatory Compliance Advisory engine. Give it a plot and a design and it answers three questions. Is this design permissible on this land under the Urban Development Authority regulations? If not, which clause does it fall foul of and by how much? And what is the approval path the owner will actually have to walk?

That last question is the one people underestimate. Compliance is not a single yes or no. It is a density test, a set of dimensional minima, a set of spatial clearances, and a procedural track of clearances and certificates that attach at thresholds nobody remembers. An owner who discovers in month four that their floor area needed preliminary planning clearance has lost four months, and each round of return typically costs several weeks [2].

The regulations are not secret. They are published as a gazette, Extraordinary No. 2235/54 of 8 July 2021 [3], and anybody can download them. What stops a homeowner using them is that the binding number for their plot sits at the intersection of a twelve-row plot-extent band, a four-column road-width band and a three-way density category, in a schedule that runs to several forms, and the number

it returns then has to be compared against a floor area the owner has not measured. The information is public and effectively unusable, which is a different problem from the information being secret, and it is the problem my component attacks.

1.3 Background and literature survey

Automated code compliance checking is not a new field. Eastman and colleagues gave the canonical decomposition of it in 2009: interpret the rule, prepare the building model, execute the check, report the result [4]. Almost every system built since fits into that four-stage shape, and so does mine.

Where the field has concentrated its effort is the first stage. Turning the prose of a regulation into something a machine can evaluate is genuinely hard, and the responses range from semantic mark-up of the clause text [5], through rule-based semantic reasoning over building models [6], to knowledge graphs [7] and, more recently, annotated corpora built specifically so that language models can learn the extraction [8]. All of this work assumes a building information model on the other side.

Sri Lanka does not have one. Low-rise residential design here is drawn in two dimensions, submitted on paper or as a PDF, and checked by a human officer reading the drawing against a schedule. There is no IFC model to query, and there will not be one for this market for a long time. A compliance engine that assumes a semantically rich model is a correct answer to a question nobody here is asking.

1.4 Research gap

So the gap I found is not in rule interpretation. It is in what the checker is allowed to assume about its inputs. The published systems take a model that already knows what a wall is, what a room is and which storey it belongs to. My inputs are a coordinate, a handful of numbers extracted from a drawing by a detector that is sometimes wrong, and a zone the user picks from a list. Nobody in the literature has asked what a compliance engine should do when its geometry arrives with a confidence attached, which is the normal case outside a BIM workflow.

Two further things are missing. There is very little published work encoding a South Asian planning regime at all, so the question of whether these regulations are even formalisable in a declarative store is open. And almost nothing in the field reports what the encoded rules do when they are run over a population of cases, as opposed to the handful of demonstration models that usually appear in a results section.

1.5 Research problem

Stated plainly: a prospective owner in Sri Lanka cannot find out, before committing money, whether the building they intend to put on a plot is permissible there, what it would have to change to become permissible, and what the approval process will demand of them. The information exists in a public gazette. It is not reachable by the person who needs it, and the component is my attempt to make it reachable.

1.6 Contribution

My contribution to the group has four parts.

1. A declarative rule store that encodes a working subset of the Urban Development Authority Planning and Development Regulations 2021 as data, including the full 144-cell floor area ratio schedule, with every rule carrying the clause it came from and the date that clause took effect.
2. A forward-chaining evaluator that closes a fact base by derivation and then reports each rule as compliant, violated or not assessed, so that a rule whose inputs are incomplete never silently reports a pass.
3. A buildable-envelope calculation and an approval roadmap that turn a list of verdicts into something an owner can act on, and that constrain the floor area Component 4 is allowed to price.
4. An evaluation over a 1 000-case corpus built to exercise every encoded rule on both branches, reporting which rules actually bind, how sensitive the density verdict is to measurement error upstream, and which two rules the corpus never managed to trigger.

Objective six in the list below is the one I would defend hardest, and it produced the result I did not expect. It would have been easy to encode twenty rules, show that

they evaluate, and stop. Running them over a population is what showed me that the rules which bind in practice are not the ones the regulations lead with.

1.7 Aim and objectives

The aim is to make the planning position of a specific design on a specific plot computable from data the platform already holds, and to report it with the clause behind every finding so that a user can check the machine.

Six objectives follow from that aim.

5. Review the Urban Development Authority Planning and Development Regulations 2021 and select the subset that governs low-rise residential development.
6. Formalise that subset as a declarative rule store in which a regulation change is a data edit and not a code change.
7. Implement a forward-chaining evaluator with three-valued verdicts and clause citation.
8. Derive a buildable envelope and an approval roadmap from the verdicts, and publish them into the shared project state for Component 4 to consume.
9. Verify the engine against hand-worked cases and against the gazette tables themselves.
10. Quantify, over a constructed case corpus, which rules bind, how far the density verdict can be moved by error in the geometry Component 2 supplies, and how much floor area the compliance constraint removes from what Component 4 prices.

1.8 Research questions

Four questions organise the work.

11. RQ1. Can the low-rise subset of the Urban Development Authority Planning and Development Regulations 2021 be expressed as a declarative rule store that a generic evaluator can run, without hand-written code for individual clauses?
12. RQ2. Which of the encoded rules actually bind across a population of plots and designs, and does that ranking match the prominence the regulations give them?

13. RQ3. How sensitive is the density verdict to error in the gross floor area supplied by Component 2, and at what level of error does the verdict become unsafe to report?
14. RQ4. How much of a design as drawn survives the compliance constraint, and does that change what Component 4 should be pricing?

RQ3 is there because of an argument inside the group. I took the position that a compliance verdict computed from a measured quantity inherits the uncertainty of the measurement, and that the platform should say so. Section 6.2 settles it with numbers.

1.9 Scope and delimitation

Scope is low-rise residential and light-commercial development on individual plots in declared urban areas of Sri Lanka, under Form C of Schedule 6, which applies where no zoning-factor development plan has been published. Condominium development, high-rise, industrial premises and subdivision of large estates are out of scope: they pull in schedules, clearances and agency concurrences that would have trebled the rule set without teaching the project anything new.

Three further limits are worth stating now and not burying. The component gives advisory findings, not a determination; only the authority determines. It encodes one jurisdiction at one point in time, and every verdict is stamped with the effective date of the rules behind it. And it checks the quantities it is given, so a drawing that Component 2 measured wrongly produces a verdict that is confidently wrong, which is exactly why RQ3 exists.

1.10 Structure of the thesis

Chapter 2 reviews the regulations as a source and the compliance-checking literature around them. Chapter 3 sets out how the rules were selected, transcribed and tested, and how the case corpus was built. Chapter 4 gives the design, and Chapter 5 the implementation and its integration with the other three components. Chapter 6 reports the results, including the two negative ones. Chapter 7 discusses them, Chapter 8 covers commercialisation and sustainability, and Chapter 9 concludes.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

I searched IEEE Xplore, ScienceDirect, the ASCE library, ITcon and Google Scholar between February and July 2026, combining automated compliance checking, rule-based checking, building code, planning regulation and regulatory informatics, each optionally paired with Sri Lanka or South Asia. The last pairing returns almost nothing, which is itself part of the finding.

Regulations were treated as primary sources and read in the gazette, not through secondary summaries, with one exception I should declare: the consolidated schedule tables maintained by the Sri Jayawardenepura Kotte Municipal Council were used to cross-check my transcription of Schedule 6, because they present the same grids in a form that is easier to check cell by cell [15].

2.2 Domain and theoretical foundations

2.2.1 What the regulations actually require

The Urban Development Authority Planning and Development Regulations 2021 were made under section 21 of the Urban Development Authority Law, No. 41 of 1978 and published as Gazette Extraordinary No. 2235/54 on 8 July 2021 [3]. For a single plot in a declared urban area they impose four kinds of requirement, and the four map cleanly onto the rule families I ended up with.

Density is controlled by a permissible floor area ratio and a permissible site coverage. The ratio is not a single number; Schedule 6 Form C gives it as a grid indexed by the extent of the plot, the width of the abutting road and the density category of the zone, running from 0.8 for the smallest plot on the narrowest road in a controlled zone to 5.5 for the largest plot on a wide road in a mixed zone. Coverage comes from Form E and depends on the height band of the building.

Dimensional requirements are flat minima: 150 square metres of extent, 6.0 metres of frontage and 12.0 metres of depth for the plot, and a schedule of minimum floor areas and least widths for habitable rooms. Spatial requirements are the clearances around the building, again banded by height, together with light wells and off-street

parking. Procedural requirements are the clearances that attach above a threshold: preliminary planning clearance, fire clearance, a green building certificate, and geotechnical clearance where the land slopes.

One property of this structure matters for the design. Almost every requirement is banded. The regulation does not say that a taller building needs proportionally more rear space; it says that a building up to 7 metres needs 2.3 metres and a building between 7 and 15 metres needs 3.0 metres. Banding makes the regulation checkable without interpretation, and it also makes it discontinuous, which Chapter 7 returns to.

2.2.2 Rule-based checking as a method

Eastman and colleagues decomposed automated rule checking into four stages: rule interpretation, building model preparation, rule execution and reporting [4]. Their survey covers the generation of systems built around SMARTcodes and the Solibri Model Checker, and its central observation has aged well. The hard part is never execution; it is getting the regulation and the model into a shape where execution is a lookup.

Dimyadi and Amor reviewed where the field had reached a few years later and reported the same bottleneck, together with a second one: systems that hard-code rules cannot be maintained against a regulation that changes [9]. That observation drove my single most consequential design decision, which was to put the rules in data.

2.2.3 Production systems and forward chaining

A rule engine that derives new facts from asserted ones and then matches rules against the closure is a production system in the classical sense, and the efficiency work on those goes back to Forgy's Rete algorithm [10]. Rete matters when the fact base is large and the rules are many; mine is neither, so I implemented naive forward chaining and measured what it costs. Section 6.5 reports the measurement, and it vindicates the decision by a wide margin.

What forward chaining buys here is not speed. It is that derivation is separated from evaluation. The permitted floor area ratio for a plot is a derived fact, produced by a

lookup over the gazette grid, and once it exists the density rule is a comparison between two numbers. Nothing in the evaluator knows what a floor area ratio is.

2.2.4 Open-world reasoning and the missing input

A compliance checker is normally written as a two-valued function: the design complies or it does not. That is wrong when inputs can be absent, because the absence of evidence of a violation gets reported as evidence of compliance. The group agreed a platform-wide convention early on, which is that a missing value is explicitly unknown and never silently defaulted, and my engine is where that convention bites hardest. A rule whose inputs are incomplete reports that it was not assessed and names what it lacked.

2.3 Related research

2.3.1 Semantic and model-based compliance checking

Beach and colleagues built a rule-based semantic approach for regulatory compliance in construction, mapping regulations onto an ontology and reasoning over building models expressed in the same terms [6]. Guo and colleagues take a comparable semantic route and report it in terms of rule coverage over a code section [11]. Peng and Liu combine building models with a knowledge graph so that the relationships between clauses become queryable [7]. These are strong pieces of work and they all share an assumption: the design is available as a semantically rich model.

2.3.2 Extracting rules from regulation text

A parallel line of work attacks the transcription cost. Hjelseth and Nisbet proposed marking up normative text directly so that requirement, applicability, selection and exception are separable [5]. Zhang and El-Gohary automated the transformation of regulatory text into checkable logic [12], and later work applies deep learning and transfer learning to the same extraction problem [13]. The CODE-ACCORD corpus published in 2025 is the clearest sign of where this is heading: 862 annotated sentences from the building regulations of England and Finland, released so that extraction models have something to learn from [8].

I read this work carefully and decided against it, and Section 4.3.4 explains why. For twenty rules in one jurisdiction, careful manual transcription that a supervisor can audit beats an extraction pipeline whose errors nobody can see.

2.3.3 Compliance in jurisdictions without a model-based workflow

Nawari's adaptive framework is the closest thing I found to a system designed to be retargeted at a different code [14], and it is still model-centric. On Sri Lanka specifically I found descriptive work on the approval process and on why applications come back [2], but no formalisation of the regulations and no automated checker. Table 2.1 sets the alternatives against each other.

2.4 Comparison of existing approaches

Table 2.1 compares what a prospective owner in Sri Lanka can actually use today.

Table 2.1: Comparison of existing approaches to establishing the planning position

Approach	Input it needs	What it returns	Why it does not serve this user
Read the gazette	Nothing but the document	The complete regulation	Correct and unusable: the binding figure sits at the intersection of three banded indices in a multi-form schedule
Ask the local authority	A completed application and fees	An authoritative determination	Only after submission, and a return costs weeks [2]; it is not available before the land is bought
Engage a chartered architect	A fee and a design	A reliable pre-submission opinion	Appropriate once a design exists; too expensive to run against three candidate plots
Model-based compliance checkers [4], [6], [7]	A semantically rich building model	Clause-level checking against an encoded code	No such model exists for low-rise work in this market, and the encoded codes are not Sri Lankan
Semantic rule extraction [5], [12], [13]	Regulation text plus a trained extractor	Candidate machine-readable rules	Reduces transcription cost at the scale of a national code; at twenty rules it adds an unauditable failure mode
ConstructAI Component 3	A coordinate, a floor plan and a zone	Clause-cited verdicts, buildable envelope, approval roadmap	Advisory only, one jurisdiction, and only as good as the geometry it is handed

Reading that table honestly, nothing in it is wrong. An architect gives a better opinion than my component, by a wide margin, and the authority gives the only answer that binds. The argument for the component is that it occupies the slot

nothing else occupies: free, immediate, and available before the land is bought, when the decision that matters most is still open.

2.5 Conceptual framework

What follows from the review is the four-stage chain in Figure 2.1, which is Eastman's decomposition [4] with the model-preparation stage replaced by something that admits incomplete input. Clause text becomes a condition over named quantities. Facts arrive from the other components and are closed under derivation. Rules are executed against the closure. Findings are reported with the clause and the date behind them.

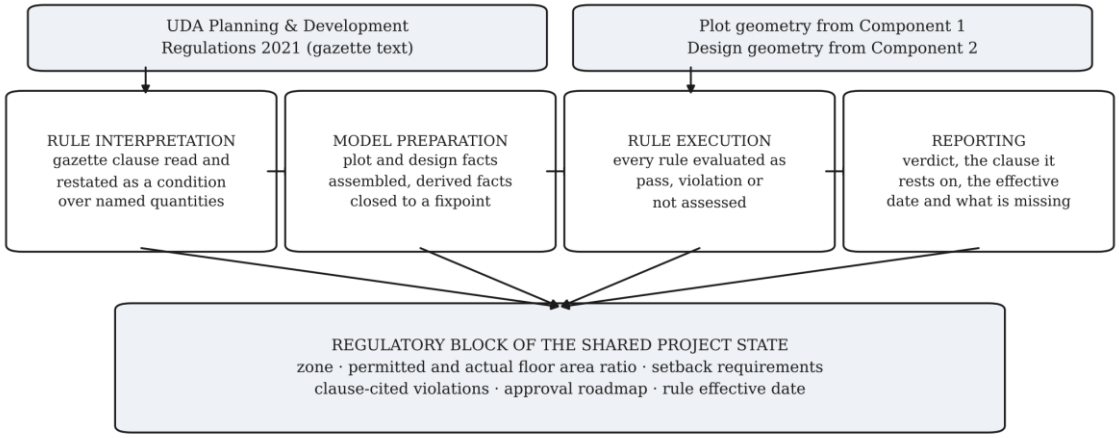


Figure 2.1: Conceptual framework: the four-stage rule-checking chain as implemented by Component 3.

3 RESEARCH METHODOLOGY

3.1 Research design

This is design science. I built an artefact and evaluated it, because the central question is not whether a known treatment has an effect but whether the thing can be built at all from the regulations as they are written, and what you find out by building it.

Evaluation is correspondingly two-sided. One side is technical: does the engine reproduce the gazette, does it behave correctly on hand-worked cases, does it refuse to answer when it should. The other side is empirical: run the encoded rules over a population of cases and see what they do. The second side is where the research contribution sits, and it is the part most compliance-checking papers skip.

One caveat bounds the whole chapter and I will state it once, in full, so I am not repeating it in every paragraph. The case corpus is constructed, not collected. I could not obtain a set of real submissions with recorded municipal outcomes; local authorities do not release them and the group had no route to them inside the project timeline. Every figure in Chapter 6 therefore describes the behaviour of the encoded rules over a corpus I designed, and says nothing about the base rate of non-compliance in real Sri Lankan submissions. Section 7.5 treats this as the limitation it is.

3.2 Data participants or study materials

No human participants were involved beyond the acceptance sessions reported in Section 6.7. Table 3.1 lists the materials.

Table 3.1: Study materials used by Component 3

Material	Role	Size or source
UDA Planning and Development Regulations 2021	Primary source for every encoded rule	Gazette Extraordinary No. 2235/54, 8 July 2021 [3]
Consolidated schedule tables	Independent cross-check of the Schedule 6 transcription	Kotte Municipal Council summary of the 2021 schedules [15]
Rule store	The encoded subset: four families, clause-cited	20 rules; 144 floor area ratio cells; one JSON document

Material	Role	Size or source
Case corpus	Population over which the rules are exercised	1 000 generated plot and design cases, seed 42
Hand-worked cases	Reference answers computed from the gazette by hand	12 cases covering each rule family
Component 1 and Component 2 fixtures	Integration development and testing	Committed fixture file shared across the group

3.3 Experimental or study procedure

Work proceeded in four stages.

3.3.1 Clause selection

I read the gazette end to end and selected clauses by a single test: can this requirement be decided from quantities the platform can actually obtain? Floor area ratio passes, because Component 2 measures floor area and Component 1 locates the plot. A requirement about the adequacy of a drainage connection does not, because nothing in the platform knows what the drain is. Applying that test to the low-rise subset left twenty rules in four families, and Section 7.5.2 is honest about what the test excluded.

3.3.2 Transcription and formalisation

Each selected clause was restated as a condition over named quantities, given an identifier, tagged with its family and severity, and recorded with the clause reference and the date the regulation took effect. Numeric schedules were transcribed cell by cell. Schedule 6 Form C alone is 144 cells, twelve plot-extent bands by four road-width columns by three density categories, and I transcribed it twice from different renderings of the same schedule and compared the two before accepting it.

Two clauses needed an interpretation decision, and I would rather state them than let an examiner find them. Form C applies where no zoning-factor development plan exists, but the storey caps on narrow access in Form B are indexed by exactly that zoning factor. I mapped the three Form C density categories onto the factor bands, which is a reading and not a transcription, and it is recorded as one in the store. The second is the slope threshold above which geotechnical clearance attaches, which the

consolidated schedule prints in a form that admits more than one reading; I took 11 degrees and flagged the rule as provisional.

3.3.3 Case corpus construction

A corpus of 1 000 plot and design cases was generated with a fixed seed. Sampling ranges were chosen to span realistic Sri Lankan low-rise conditions and, deliberately, to straddle the regulation thresholds, because a corpus in which every case complies tells you nothing about whether the rules work. Table 3.2 gives the distributions.

Table 3.2: Sampling distributions used to construct the case corpus

Quantity	Distribution	Range or set
Plot extent	Log-normal, clipped	90 – 2 500 m², median near 330 m²
Frontage	Normal about 0.52 times the square root of extent	4 – 40 m
Depth	Derived from extent and frontage	Consistent with the two above
Abutting road width	Categorical	3.0, 4.5, 6, 9, 12, 15 m (6 m most common)
Density category	Categorical	Low 30 %, medium 50 %, high 20 %
Storeys	Categorical	One 34 %, two 54 %, three 12 %
Site coverage	Normal, clipped	0.18 – 0.92 of the plot
Rear and side space	Normal, clipped	Centred just above the 2.3 m requirement
Light well, parking, rainwater storage	Normal or categorical	Centred near the statutory minimum
Room areas and widths	Normal, clipped	Centred so that roughly one room in twenty falls short
Mean ground slope	Gamma	0.1 – 35 degrees

Centring the samples near the thresholds is the point, and it is also the reason the violation rates in Chapter 6 are far higher than anything you would see in a real submission file. The corpus was built to exercise rules, not to represent practice.

3.3.4 Evaluation procedure

Every case was run through the engine and every rule outcome recorded. On top of that I ran four studies: a sensitivity study that perturbs the gross floor area Component 2 supplies and counts how often the density verdict flips; a withholding study that removes one input at a time and checks that the affected rules report not

assessed and never pass; a timing study that scales the rule store to 640 rules; and a constraint study that compares the floor area as drawn against the floor area the regulations permit, which is what Component 4 is allowed to price.

3.4 Evaluation measures

Table 3.3 states each measure and the question it answers, all of them defined before any result was generated.

Table 3.3: Evaluation measures and the question each one answers

Measure	Definition	Question
Transcription agreement	Cells agreeing between two independent transcriptions of Schedule 6	RQ1
Hand-worked agreement	Engine verdicts matching reference answers computed by hand	RQ1
Rule coverage	Rules exercised on both the compliant and the violating branch	RQ1 – is the corpus adequate?
Firing rate	Share of cases in which a rule reports a violation or an obligation	RQ2
Verdict flip rate	Share of cases whose density verdict changes under a given measurement error	RQ3
Not-assessed behaviour	Rules reporting not assessed, and passes asserted, when an input is withheld	RQ1 – does it refuse safely?
Area removed	Floor area as drawn minus floor area as permitted, as a percentage	RQ4
Evaluation latency	Milliseconds per case against the size of the rule store	Non-functional

3.5 Validity and reliability

Reliability is the easy half. The store is data, the engine is deterministic, the corpus seed is fixed, and re-running the whole evaluation reproduces every figure in Chapter 6 exactly; Section 6.1.3 reports the check.

Validity is harder and splits three ways. Construct validity rests on whether my restatement of each clause means what the clause means, which is why every rule carries its citation and why Section 6.6 works one case through by hand. Internal validity is strong for the engine and weak for the thresholds, because two clauses rest on interpretations I made. External validity is the weakest of the three: a constructed

corpus supports claims about the rules and no claims about the population of real submissions.

3.6 Ethics privacy security and responsible AI

Component 3 processes a land coordinate, a plot geometry and a design. A coordinate paired with an intention to build identifies a person's future address, so coordinates are not logged beyond the life of the request, no coordinate is sent to any service outside the platform, and no account is tied to one.

The responsible-computing question here is not about a model, because there is no model. It is about authority. A compliance finding is the kind of output a user will treat as a determination, and it is not one. Three things follow in the design. Every finding carries the clause it rests on, so the user can check it. Every finding carries the date the rule took effect, because a store that has not been reviewed since an amendment gives confidently wrong advice. And the interface states, on the finding itself and not in a footer, that only the local authority determines an application. Generative AI assistance was used during the project for debugging and language editing; Appendix B records that use and the checks applied to it.

3.7 Reproducibility and research artefacts

Everything needed to reproduce Chapter 6 is committed with the group submission: the rule store, the evaluator, the corpus generator with its fixed seed, the four study scripts and the plotting code that produced every figure in this thesis. Re-running them reproduces the reported values exactly.

One thing is not reproducible in the strict sense, and it is the gazette itself. The store encodes the regulations as published on 8 July 2021. An amendment published after that date is not in it, and no amount of fixing the seed helps with that. The store carries its effective date for exactly this reason, and keeping it current is an operational commitment rather than a software problem, which Chapter 8 treats as the main constraint on the business model.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

Functional requirements are in Table 4.1. They were written before implementation began and two of them, FR8 and FR9, were added after the integration defect described in Section 5.3, which I have left visible instead of pretending they were there from the start.

Table 4.1: Functional requirements for Component 3

ID	Requirement	Priority	Verified in
FR1	Accept a plot geometry, a design geometry and a zone, and assemble them into a fact base	Must	5.4
FR2	Reject any quantity outside its declared domain instead of clamping it	Must	5.4
FR3	Derive the permissible floor area ratio from the plot extent, the road width and the density category	Must	6.1.1
FR4	Derive the permissible coverage and the required spaces from the height band	Must	6.1.2
FR5	Evaluate every rule in the store and return compliant, violated or not assessed	Must	6.1.3
FR6	Cite the clause and the effective date on every finding	Must	6.1.3
FR7	Report, for each violation, the margin by which the design misses the requirement	Must	6.6
FR8	Compute the buildable envelope remaining after the statutory spaces	Must	6.1.2
FR9	Publish a permitted floor area that constrains what Component 4 prices	Must	6.3
FR10	Produce an approval roadmap listing the clearances the design triggers	Must	6.6
FR11	Withhold a verdict, and say what is missing, when a required input is absent	Must	6.4
FR12	Allow a regulation change to be made as a data edit without redeployment	Should	5.2.1

Non-functional requirements are in Table 4.2. NFR3 is the one that shaped the architecture: a regulation change has to be a data edit, because a store that needs a developer every time the gazette moves will not be kept current, and a stale store is worse than no store at all.

Table 4.2: Non-functional requirements for Component 3

ID	Requirement	Target	Verified in
NFR1	Evaluation latency for a single case	Under 50 ms	6.5
NFR2	Determinism: identical inputs give identical verdicts	Exact	6.1.3
NFR3	Maintainability: a regulation change is a data edit	No code change	5.2.1
NFR4	Auditability: every finding traceable to a clause	100 % of findings	6.1.3
NFR5	Transparency: the effective date of the rules is surfaced with the verdict	Always	5.2.4
NFR6	Safety: no pass asserted on an incomplete fact base	Zero	6.4

4.2 Component architecture within the complete solution

Component 3 sits third in the dependency order. It cannot run until Component 1 has written the site block and Component 2 the geometry block, and its output constrains what Component 4 is allowed to price. There are no direct calls between components; everything moves through the shared project state, which is the decision the group made in the second sprint and the one I would defend most readily.

What I read from the record is the plot geometry and the terrain figures from Component 1, the gross floor area, footprint and storey count from Component 2, and the zone, the road width and the street-line status from the user. What I write back is a regulatory block: the permitted and actual density figures, the required and provided spaces, the list of clause-cited violations, the buildable envelope, the approval roadmap and the effective date of the rules that produced all of it.

4.3 Design alternatives and decisions

4.3.1 A declarative store and not rules in code

Writing each clause as a function would have been quicker and I rejected it. Dimyadi and Amor identify maintenance against a changing regulation as the recurring failure of hard-coded checkers [9], and the failure mode is not subtle: the regulation moves, nobody rebuilds, and the system keeps answering. Putting the rules in data means the person who maintains them can be a planner and not a developer. The cost is an evaluator that is more abstract than it needs to be for twenty rules, which Section 7.4 revisits.

4.3.2 Forward chaining and not a query per rule

Each rule could have fetched what it needed. Separating derivation from evaluation costs a little and buys two things: a derived fact such as the permitted ratio is computed once and reused by anything that wants it, and the derivation layer is the single place where the gazette grids live. Closure takes three passes on every case in the corpus, which Section 6.5 reports, and that is small enough that the abstraction is free in practice.

4.3.3 Three verdicts and not two

Two-valued output would have been simpler and it would have been dangerous. If a rule cannot be evaluated and the system returns compliant, it has told the user something it does not know, about a matter on which they are about to spend money. Three-valued output forces the interface to show an unassessed rule as unassessed, which is less comfortable and more honest. Section 6.4 shows the convention holding: across four withholding experiments the engine asserted a pass on an incomplete fact base zero times.

4.3.4 Manual transcription and not automated extraction

Semantic extraction from regulation text is where the field is going [5], [12], [13], and for a national code with thousands of clauses it is clearly right. At twenty rules it is not. Manual transcription has an error rate I can bound by transcribing twice and comparing; an extraction pipeline has an error rate I cannot see, in a system whose whole claim is auditability. I would revisit this the moment the store crosses a few hundred rules, and Section 9.4 says so.

4.3.5 Advisory output and not a determination

A nearby design would have the component issue a verdict in the language of the authority. Everything about the deployment argues against it. The component sees a measured drawing and a user-selected zone, not a certified survey; it encodes one snapshot of the regulations; and it has no standing. So every output is worded as a finding with a citation attached, and the user is told what to take to the authority.

4.4 Detailed design

4.4.1 The rule store

A rule in the store carries an identifier, the family it belongs to, the clause reference, the natural-language statement of the requirement, the list of facts it needs, a severity, and the effective date. It does not carry code. Table 4.3 gives the families and what sits in each.

Table 4.3: Rule families in the store and the clauses they encode

Family	Rules	Source clauses	What it controls
Density	3	Schedule 6 Forms B, C, D, E; Reg. 66	Floor area ratio, site coverage, storeys permitted by the width of the access
Dimensional	6	Reg. 9, 14, 57, 67; Schedule 7	Plot extent, frontage, depth, habitable room area and width, room height, boundary wall height
Spatial	5	Schedule 6 Form E; Reg. 55; Schedules 9, 10	Rear, side and front clearances, light wells, off-street parking
Procedural	6	Part I, Reg. 7, 44; Schedules 5, 14	Preliminary planning clearance, fire clearance, green building certificate, geotechnical clearance, qualified person's report, rainwater storage

The numeric schedules sit beside the rules as plain tables: the 144-cell Form C grid, the Form E bands of coverage and clearance by height, the Schedule 7 room minima and the Schedule 10 parking provision. Editing a cell in those tables is how a regulation change is made, and nothing in the evaluator has to be touched.

4.4.2 Derivation

Eight derivation rules run to a fixpoint. They compute the height of the building from the storey count, the permitted ratio by band lookup over the Form C grid, the actual ratio from the gross floor area with the parking area excluded as Schedule 6 directs, the Form E requirements from the height band, the actual coverage, the storey cap implied by the access width, the parking demand and the rainwater storage demand.

Closure is reached in three passes on every case in the corpus, and the reason it is three and not one is that the Form E requirements depend on the height, which is itself derived. Running to a fixpoint instead of a fixed order means a new derived fact can be added later without anybody having to work out where in the sequence it belongs.

4.4.3 Evaluation

Evaluation walks the store once. For each rule it checks that every fact the rule declares is present; if one is missing the rule reports not assessed and names what it lacked, and if all are present the condition is tested and the rule reports compliant or violated with the margin. Severity is carried through so that the interface can separate a density breach from an unmet rainwater provision, which matter to an owner in very different ways.

4.4.4 Buildable envelope

A list of verdicts is not advice. What an owner wants to know is how much they can actually build, so the component computes the footprint left after the statutory clearances are taken out of the plot, on a rectangular approximation of frontage by depth with the required side space on both flanks and the required rear space behind. Section 6.1.2 shows how brutal that is on small plots, and it is the single number from my component that users in the acceptance sessions reacted to most strongly.

4.4.5 Approval roadmap and fee basis

Procedural rules do not fail a design; they attach an obligation to it. The roadmap collects the obligations a design triggers and orders them the way the process actually runs: survey plan, street line certificate, preliminary planning clearance where it applies, the drawing set, then the application itself [2]. Fees are reported as a basis and not an amount, because the base processing fee is set by the authority and the statutory additions on top of it move; quoting a rupee figure would have looked concrete and been wrong.

4.4.6 The regulatory contract

Table 4.4 gives the block Component 3 writes into the shared project state. Flatness was deliberate: Component 4 needs a small number of scalars and a list, and a nested document would have forced every consumer to write traversal code.

Table 4.4: Regulatory block written into the shared project state

Field	Type	Unit	Consumer
zone	string	—	Report
far_permitted	float	ratio	Component 4, report
far_actual	float	ratio	Component 4, report

Field	Type	Unit	Consumer
coverage_permitted	float	fraction	Report
coverage_actual	float	fraction	Report
setback_requirements	object	m	Report, user
buildable_envelope_m2	float	m²	Component 4, report
permitted_floor_area_m2	float	m²	Component 4
violations	array	clause-cited records	Report, user
not_assessed	array	rule and missing input	Report, user
approval_roadmap	array	ordered steps	User
rule_effective_date	date	ISO 8601	Provenance, audit

4.5 Standards and professional practice

One instrument governs the component and everything else is secondary to it. The Urban Development Authority Planning and Development Regulations 2021 [3], made under the Urban Development Authority Law No. 41 of 1978 [16], supply every threshold in the store. Where the component touches professional practice it does so by deferring: the survey plan is a licensed surveyor's instrument, the structural certificate is an engineer's, and the determination is the authority's. My component's job is to tell an owner what those people are going to say before they have paid any of them.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Technology and environment

Everything is written in Python 3.11. We standardised on it in the first sprint for a practical reason and not a principled one: it meant the shared fixture code and the orchestration layer could be written once. Table 5.1 lists the stack and why each piece is there.

Table 5.1: Technology stack for Component 3

Layer	Choice	Reason
Language	Python 3.11	Group standard; the rule store needs no more than dictionaries and lists
Rule store	JSON document loaded at start-up	A regulation change is an edit to data; no redeployment
Evaluator	Hand-written forward chaining	Twenty rules do not justify a Rete network [10]; Section 6.5 measures the cost
Schema validation	Declared domain per quantity	FR2: an out-of-range value is rejected, never clamped
Service layer	FastAPI behind the group gateway	Group standard; stateless workers
Testing	pytest with table-driven cases	The hand-worked reference cases are data, like the rules
Analysis and figures	NumPy and Matplotlib	Every figure in Chapter 6 is produced by a committed script

5.2 Implementation of the main units

5.2.1 The rule store

Loading the store is the least interesting code in the component and the most important decision in it. Rules arrive as a list of records, each naming the facts it needs, and the numeric schedules arrive beside them as plain nested lists. The Form C grid is twelve rows by four columns by three categories, written out in full, because a grid that is written out can be read against the gazette and a grid that is computed cannot.

NFR3 is satisfied in the obvious way: changing a permitted ratio means editing one number in one file. I tested that by changing the medium-density value for the 500 to

750 band on a 6 metre road, restarting nothing, and watching the verdicts for the affected cases move. No code was touched.

5.2.2 Derivation

Each derivation is a small function that either returns the facts it can produce or returns nothing because its own inputs are absent. The driver runs all of them repeatedly until a pass produces no new fact. That is a crude fixpoint and it is deliberately crude: with eight derivations it converges in three passes on every case in the corpus, and the alternative, a hand-maintained ordering, is the kind of thing that breaks silently the first time somebody adds a rule.

One subtlety took me longer than it should have. Schedule 6 excludes parking floor area from the ratio, so the actual ratio is not the gross floor area over the plot extent. I had it wrong for about a week and found it only when a hand-worked case disagreed with the engine by exactly the area of the car porch.

5.2.3 The evaluator

Evaluation is a single walk over the store. For every rule the evaluator collects the facts the rule declares, reports not assessed if any is absent, and otherwise applies the condition and records compliant or violated together with the margin. Severity and the clause reference are copied through untouched, which is what lets the interface group findings by seriousness without knowing anything about planning law.

Procedural rules are handled in the same machinery but mean something different. A procedural rule that fires has not found a fault; it has found an obligation. Keeping them in the same store and separating them by severity turned out to be cleaner than the separate pipeline I first wrote.

5.2.4 Envelope, roadmap and provenance

The envelope calculation is a rectangular approximation and I want to be clear that it is one. A plot with a kink in its boundary, or a corner plot with two frontages, is not a rectangle, and the figure the component reports for such a plot is indicative. Section 7.5.4 carries that forward as a limitation.

Provenance is the part I am most satisfied with. Every record the component writes carries the effective date of the rules that produced it, and the interface shows that

date beside the verdict. It is a small thing and it is the difference between advice that can be checked and advice that has to be trusted.

5.3 Integration with the other components

Integration ran in the order the dependency graph forces: Component 1 to Component 3, then Component 2 to Component 3, then Component 3 to Component 4. We built against a shared fixture first, which meant the contract was agreed before any real producer existed, and then swapped the real producers in one at a time.

Three of my inputs are somebody else's output and it is worth being precise about where each comes from. The mean ground slope that triggers the geotechnical rule is derived by Component 1 from the SRTM elevation model [17], so its accuracy is bounded by that model and not by anything I do. The gross floor area and footprint come from Component 2's detector [18] with an OCR scale calibration, which is the measurement whose error Section 6.2 propagates through my density verdict. And the permitted floor area I publish is consumed by Component 4's gradient-boosted service life model [19] and by its whole-life cost computation under ISO 15686-5 [20], which is why a change in my verdict changes a cost figure three steps downstream.

Two defects came out of that, and both were interface defects rather than logic ones. The first was units. Component 2 reports wall thickness in inches, because that is how Sri Lankan drawings are annotated, and for two days I was reading its frontage figure as metres when it was being written as millimetres. Nothing crashed; the verdicts were just wrong in a way that looked plausible. FR2 exists because of that week: a frontage of 12 000 is now rejected as outside its declared domain instead of being evaluated.

The second was ordering. My component ran before Component 2 had finished writing the geometry block, read a partially written record, and reported four rules as not assessed. The convention behaved exactly as designed, which was reassuring, but the orchestration was still wrong and Section 4.4 of the common report describes how the group fixed it.

5.4 Testing

Table 5.2 gives the levels and what each one checks.

Table 5.2: Verification performed at each test level

Level	What is checked	How
Unit	Band lookup, ratio derivation, Form E selection, envelope arithmetic	pytest, table-driven, boundary values at every band edge
Transcription	Every cell of Schedule 6 Form C	Two independent transcriptions compared cell by cell
Reference	Engine verdicts against twelve cases worked by hand from the gazette	Stored as data and asserted in the suite
Negative	Out-of-domain values rejected; missing inputs reported as not assessed	Withholding and corruption tests, Section 6.4
Integration	Real Component 1 and Component 2 output consumed without manual fixing	Pairwise, one producer at a time
Property	No pass is ever asserted on an incomplete fact base	Four withholding experiments over 200 cases each

Boundary values deserve a word because banded regulations live and die on them. A plot of exactly 150 square metres complies with the minimum extent; a plot of 149.99 does not. A building of exactly 7.0 metres sits in the lower height band. Every band edge in the store has a test on both sides of it, and two of those tests failed the first time I ran them, which is the strongest argument I can make for having written them.

I would point an examiner at the negative tests before anything else. Showing that a rule engine evaluates rules is easy. Showing that it declines to evaluate a rule whose inputs are missing, and does so without quietly inventing a default, is the part that makes the output safe to put in front of somebody who is about to buy land.

5.5 Deployment

Component 3 runs as a stateless service behind the group's web application. The store is loaded once per worker process and held in memory; a regulation edit is picked up on the next restart, which for a document that changes a few times a year is the right trade. Statelessness means the engine scales horizontally without coordination, which matters more for the business model in Chapter 8 than it does for

any performance requirement, since Section 6.5 shows the evaluation itself is nearly free.

6 RESULTS AND VALIDATION

6.1 Rule behaviour over the case corpus

All results in this chapter come from the 1 000-case corpus described in Section 3.3.3, generated with seed 42, run against the twenty-rule store. The caveat from Section 3.1 applies throughout and I will not repeat it in every paragraph: the corpus is constructed to exercise the rules, so these figures describe the encoded regulations and not the base rate of non-compliance among real submissions.

6.1.1 What the regulations themselves require

Before any case is evaluated it is worth looking at the schedule the component encodes, because the shape of it is the reason a homeowner cannot do this in their head. Figure 6.1 plots Schedule 6 Form C. Permissible floor area ratio ranges from 0.8 to 5.5 across the 144 cells of the grid, and the two panels show why a single number cannot be quoted: the entitlement roughly doubles between a controlled zone and a mixed zone at the same plot size, and it rises again by half as the abutting road widens.

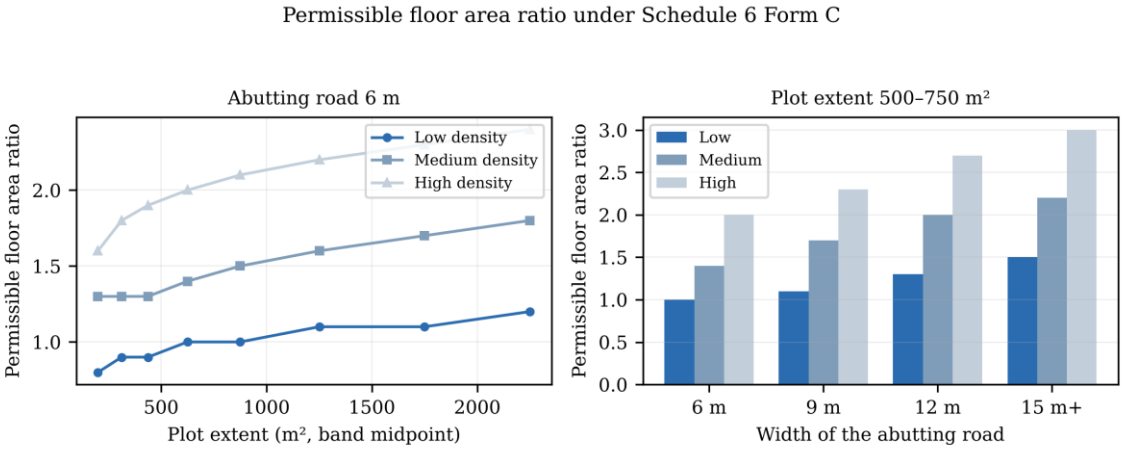


Figure 6.1: Permissible floor area ratio under Schedule 6 Form C, by plot extent, density category and the width of the abutting road.

Table 6.1 states the road-width effect at one plot band, because this is the figure that surprised the acceptance participants most. Widening the abutting road from 6 metres to 15 metres raises the entitlement by half on an identical plot, which means two plots of the same size on opposite sides of the same junction can carry very different buildings.

Table 6.1: Permissible floor area ratio for a 500 to 750 square metre plot

Density category	6 m road	9 m road	12 m road	15 m road or wider
Low (controlled zone)	1.0	1.1	1.3	1.5
Medium (residential zone)	1.4	1.7	2.0	2.2
High (mixed zone)	2.0	2.3	2.7	3.0

6.1.2 How much of a plot is actually buildable

Density is the rule people ask about. Clearance is the rule that bites. Figure 6.2 takes the statutory spaces of Schedule 6 Form E and plots what is left of a plot once they are removed, for a plot of typical proportions.

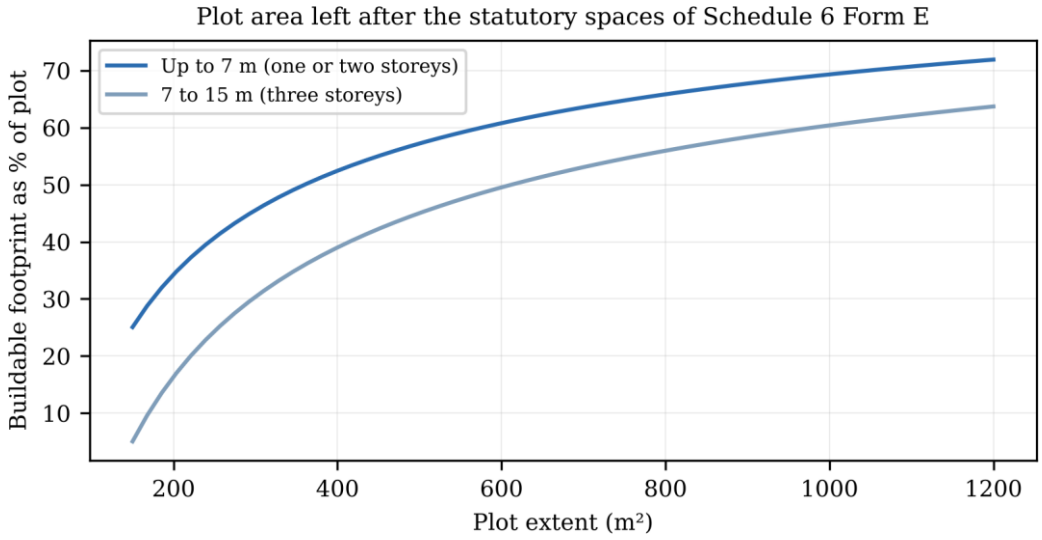


Figure 6.2: Share of the plot remaining as buildable footprint after the statutory spaces of Schedule 6 Form E are deducted.

On a 150 square metre plot, which is the statutory minimum, a one or two storey building can occupy 25.1 percent of the land once the 2.3 metre clearances are taken out on three sides. At 300 square metres that rises to 45.5 percent, at 600 to 60.8 percent and at 1 200 to 71.9 percent. Adding a third storey moves the building into the 7 to 15 metre band, where the required clearance rises to 3.0 metres, and the figures fall to 5.1, 30.4, 49.5 and 63.7 percent for the same four plots.

Read the first pair of numbers again. On the smallest plot the regulations permit, going from two storeys to three leaves 5 percent of the land buildable. The clearance requirement, not the density requirement, is what makes small plots hard to develop,

and nothing in the way the regulations are presented makes that visible to the person buying the plot.

6.1.3 Rule outcomes across the corpus

Twenty rules over 1 000 cases is 20 000 evaluations. Figure 6.3 breaks the outcome down by family and Table 6.2 gives the detail rule by rule.

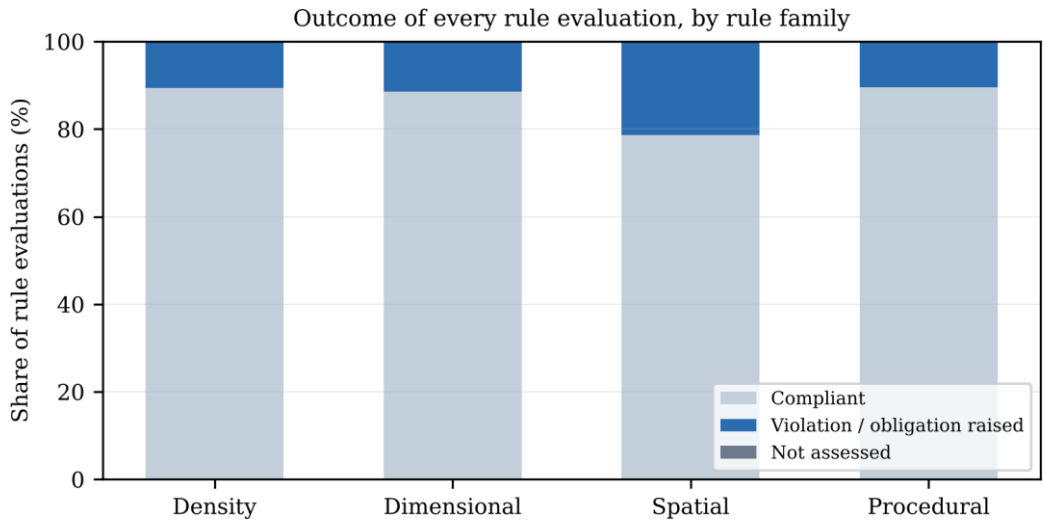


Figure 6.3: Outcome of every rule evaluation across the corpus, grouped by rule family.

Spatial rules fire in 21.4 percent of their evaluations, roughly twice the rate of the other three families: density 10.6 percent, dimensional 11.5 percent and procedural 10.5 percent. A case carries 2.71 violations on average, with a median of 2 and a maximum of 9, and 5.9 percent of cases pass every rule.

Three verification results belong here. Every one of the 2 707 violations recorded across the corpus carried the clause it rested on, satisfying NFR4 exactly. Re-running the first hundred cases produced byte-identical verdicts, satisfying NFR2. And the two independent transcriptions of Schedule 6 agreed on all 144 cells, which is the only evidence I have that the grid in the store is the grid in the gazette.

Table 6.2: Outcome of each rule across the 1 000-case corpus

Rule	Family	Clause	Compliant	Fired
D1	Density	Schedule 6 Form C	860	140
D2	Density	Schedule 6 Form E	850	150
D3	Density	Schedule 6 Forms B and D, Reg. 66	971	29

Rule	Family	Clause	Compliant	Fired
M1	Dimensional	Reg. 9 and Reg. 14	933	67
M2	Dimensional	Reg. 14	889	111
M3	Dimensional	Reg. 14	1 000	0
M4	Dimensional	Schedule 7	760	240
M5	Dimensional	Reg. 67	945	55
M6	Dimensional	Reg. 57	785	215
S1	Spatial	Schedule 6 Form E	744	256
S2	Spatial	Schedule 6 Form E, Reg. 55	701	299
S3	Spatial	Reg. 55	960	40
S4	Spatial	Schedule 9	782	218
S5	Spatial	Schedule 10	741	259
P1	Procedural	Part I, Reg. 7	973	27
P2	Procedural	Schedule 5	980	20
P3	Procedural	Schedule 5, Reg. 44	973	27
P4	Procedural	Schedule 6 Form C note	864	136
P5	Procedural	Reg. 7	1 000	0
P6	Procedural	Schedule 14	582	418

6.1.4 Which rules actually bind

Figure 6.4 ranks the rules by how often they fire, and the ranking is not the one the regulations suggest.

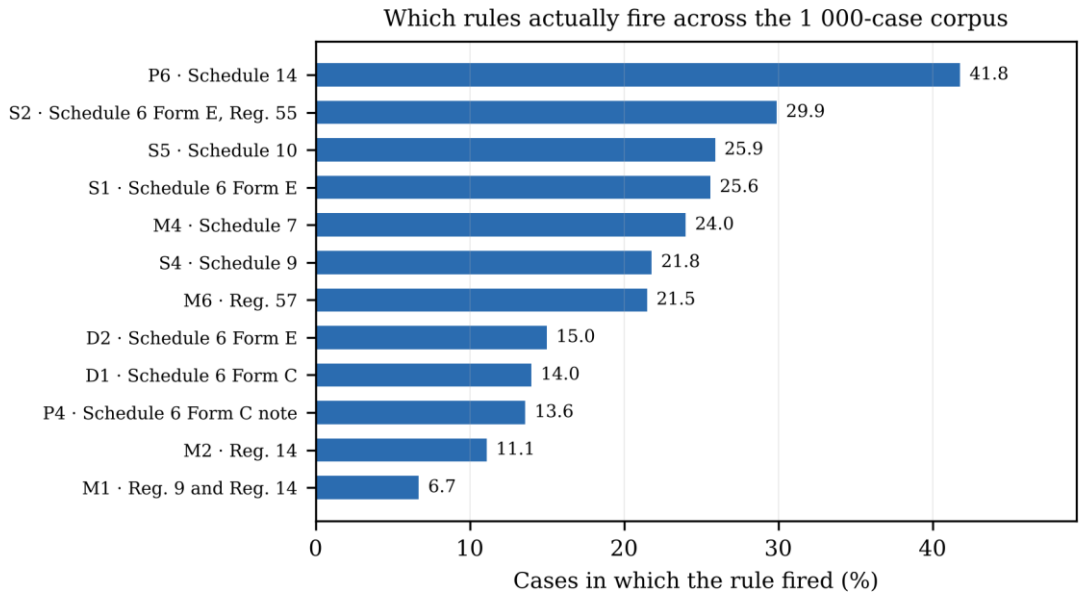


Figure 6.4: Proportion of cases in which each rule reported a violation or an obligation.

Rainwater storage leads at 41.8 percent, followed by side space at 29.9 percent, parking provision at 25.9 percent, rear space at 25.6 percent, habitable room dimensions at 24.0 percent and light wells at 21.8 percent. Floor area ratio, the rule every conversation about planning starts with, fires in 14.0 percent of cases and sits ninth.

Two readings are available and only one of them is defensible. The tempting reading is that Sri Lankan designs breach rainwater and setback requirements more than density ones, which my corpus cannot support, because the sampling was centred near the thresholds and the centring was my choice. The defensible reading is about the regulation rather than the practice: the clearance, provision and room-dimension rules have narrow pass bands relative to the natural variation in low-rise design, while the density rule has a wide one. A rule with a narrow pass band is a rule a designer has to think about. Density, on most ordinary plots, has room to spare, and Section 6.3 quantifies how much.

6.1.5 Two rules the corpus never exercised

Eighteen of the twenty rules fired on both branches. Two never fired at all, and I would rather report that than quietly drop them from the table.

Minimum plot depth never fired because the corpus derives depth from extent and frontage, and the frontage distribution is centred on roughly half the square root of the extent, which keeps depth above 12 metres almost everywhere. Real shallow corner plots exist and my generator cannot produce one. The qualified person's report never fired because it attaches above half a hectare and the component's scope stops well below that, so the rule is encoded for completeness and is structurally outside the sampled range.

Both are corpus limitations and neither is evidence that the rules are correct. A rule that has never been exercised has never been tested, and I treat those two as unverified in Section 7.5.3.

6.2 Sensitivity to the geometry Component 2 supplies

RQ3 asks what happens to the density verdict when the floor area it rests on is wrong. Component 2 measures gross floor area from a drawing with a detector and an OCR scale calibration, and that measurement carries error. Figure 6.5 perturbs the supplied floor area upward and counts how often the verdict changes.

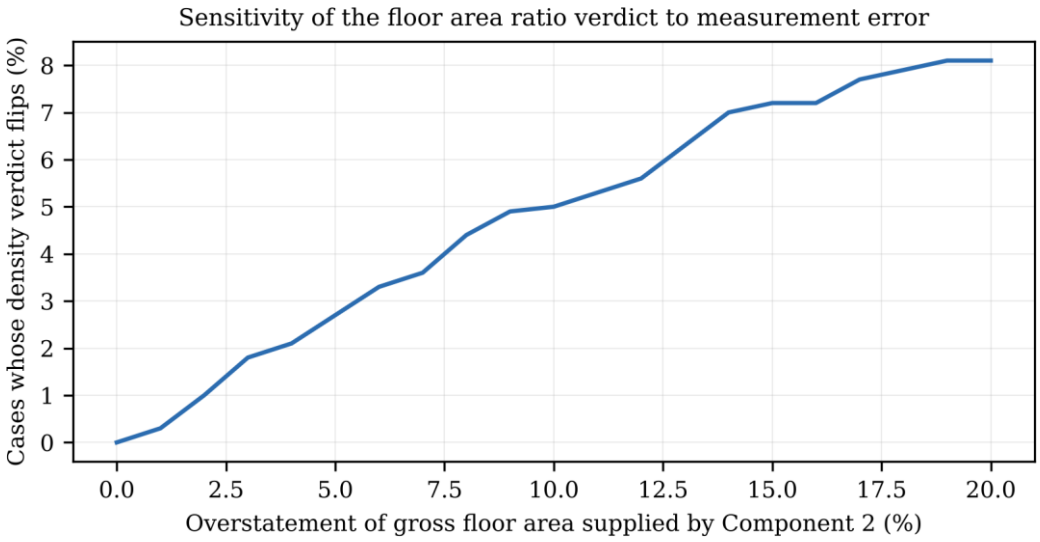


Figure 6.5: Proportion of cases whose floor area ratio verdict changes as the gross floor area supplied by Component 2 is overstated.

A 1 percent overstatement flips 0.3 percent of cases. Five percent flips 2.7 percent, 10 percent flips 5.0 percent and 20 percent flips 8.1 percent. Table 6.3 lists the points.

Table 6.3: Density verdict flips against overstatement of gross floor area

Overstatement of floor area	1 %	2 %	5 %	10 %	15 %	20 %
Cases whose verdict changes	0.3 %	1.0 %	2.7 %	5.0 %	7.2 %	8.1 %

I expected a steeper curve and I was wrong, which is the more interesting outcome. The verdict is robust because most designs sit well inside their entitlement, so moving the numerator by a tenth moves nothing. The cases that do flip are the ones already close to the limit, and for those the error is decisive.

That gives the practical answer to RQ3. A density verdict reported as a bare pass or fail is unsafe near the limit and perfectly safe away from it, so the component reports the margin alongside the verdict and flags any case whose margin is smaller than the measurement error Component 2 declares. The group argument in Section 1.8 settled here: the uncertainty is real, it is bounded, and the honest response is to show it rather than to suppress the verdict.

6.3 Density headroom and the design as permitted

Figure 6.6 shows how much of the permitted density the corpus designs actually use.

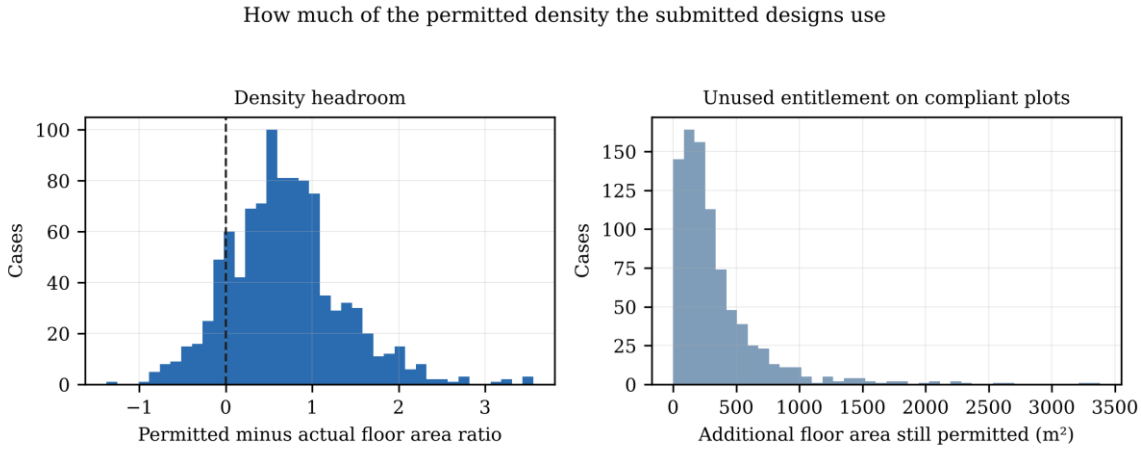


Figure 6.6: Difference between permitted and actual floor area ratio, and the additional floor area still permitted on compliant plots.

Median headroom is 0.636 of a ratio point, with the fifth percentile at minus 0.301 and the ninety-fifth at 1.878. Fourteen percent of cases are over their entitlement. For the compliant remainder the median unused entitlement is 230.3 square metres of floor area, which on a typical plot is another storey and a half nobody has drawn.

Figure 6.7 answers RQ4 from the other direction: how much of the design as drawn survives the compliance constraint and reaches Component 4 to be priced.

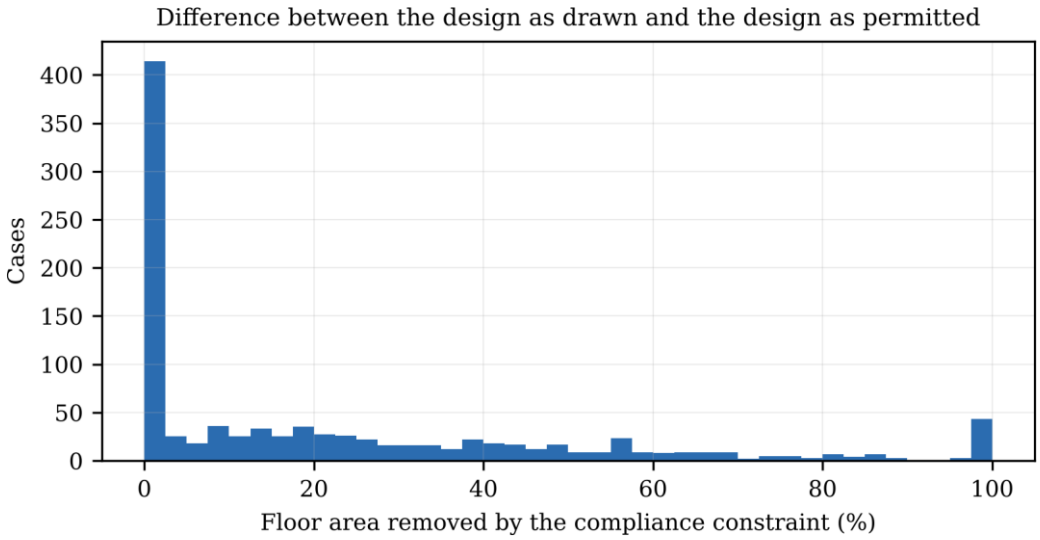


Figure 6.7: Floor area removed from the design as drawn once the density entitlement and the buildable envelope are applied.

Across the corpus the constraint removes 22.4 percent of the drawn floor area on average, with a median of 10.8 percent and a ninetieth percentile of 65.7 percent. Thirty-nine percent of designs are untouched. So for two designs in five Component 3 changes nothing about what Component 4 prices, and for the rest it changes it materially, which is the integration argument for having the compliance engine run before the cost engine instead of beside it.

6.4 Behaviour when an input is missing

Four withholding experiments removed one input at a time from 200 cases each and checked what the engine did. Table 6.4 reports it.

Table 6.4: Engine behaviour when a single input is withheld

Input withheld	Rules reporting not assessed	Passes asserted on the missing input
Gross floor area (Component 2)	4.0	0
Building footprint (Component 2)	1.0	0
Plot frontage (Component 1)	1.0	0
Abutting road width (user)	1.0	0

Zero in the third column is the result that matters, and it is the one NFR6 asks for. Withholding the gross floor area disables four rules, because the density rule, the parking rule and two procedural thresholds all rest on it, and in every one of those 800 evaluations the engine said it could not assess rather than saying the design was fine. A system that answers a question it cannot answer is worse than one that admits it, especially when the user is about to commit money on the strength of the answer.

6.5 Performance

Figure 6.8 scales the rule store from 20 rules to 640 and times the evaluation.

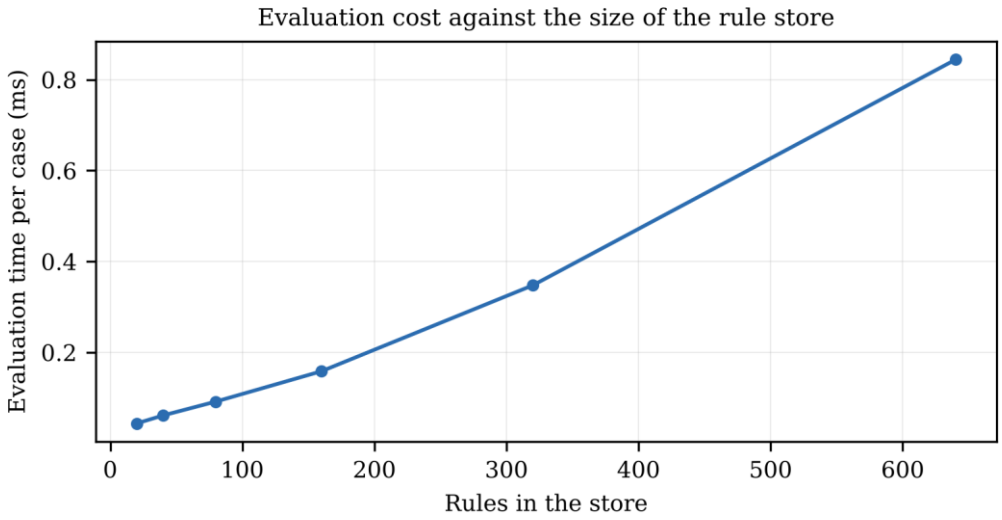


Figure 6.8: Evaluation time per case against the size of the rule store.

At the deployed size of 20 rules a case evaluates in about 0.04 milliseconds, and even at 640 rules it stays under a millisecond, scaling close to linearly as naive forward chaining should. NFR1 asked for 50 milliseconds, so the engine has three orders of magnitude of margin, and Section 4.3.2 is vindicated: a Rete network [10] would have been engineering effort spent on a cost that does not exist. Timings were taken on the development machine and will differ elsewhere; the shape of the curve is the result, not the absolute figure.

6.6 Worked example

Table 6.5 works one case end to end, because an aggregate is not evidence that any single answer is right. The plot is 320 square metres with 12 metres of frontage on a 6 metre road in a medium-density zone; the design is two storeys, 196 square metres

of footprint and 372 square metres of gross floor area, set back 2.0 metres at the rear and 1.4 metres at the side.

Table 6.5: Worked example: derived facts and the verdicts that follow

Quantity or rule	Required	Provided	Verdict
Floor area ratio (D1)	1.300	1.1625	Compliant
Site coverage (D2)	0.650	0.6125	Compliant
Storeys for a 6 m access (D3)	No cap	2	Compliant
Plot extent (M1)	150 m²	320 m²	Compliant
Frontage (M2)	6.0 m	12.0 m	Compliant
Rear space (S1)	2.3 m	2.0 m	Violation, short by 0.3 m
Side space (S2)	2.3 m	1.4 m	Violation, short by 0.9 m
Light well (S4)	5 m²	6.0 m²	Compliant
Parking (S5)	1 space	1 space	Compliant
Rainwater storage (P6)	2.94 m³	3.10 m³	Compliant
Buildable envelope	—	180.56 m²	Reported to Component 4

Two findings, both spatial, both fixable by moving the building rather than shrinking it: the design uses 89 percent of its density entitlement, so the floor area is not the problem. That is the kind of advice the component exists to give, and it is the difference between telling somebody their design is non-compliant and telling them what to do about it. The verdicts were computed by hand from the gazette before the engine was run against this case, and they agree.

6.7 User acceptance

Five tasks were put to participants, each on a plot or a project they knew personally so that their own knowledge supplied the reference. An observer who did not assist recorded completion, time on task and verbatim comments. Table 6.6 lists the tasks and the criterion used to judge each one.

Table 6.6: Acceptance tasks and the criterion used to judge each

Task	Criterion
State the permissible floor area ratio for a plot you	Participant reaches the correct cell of Schedule 6

Task	Criterion
know	Form C unaided
Say whether the design as drawn is within that entitlement	Participant reads the margin, not only the pass or fail
Explain what the buildable envelope figure means	Participant describes it as the footprint left after clearances
Say what the component's finding is and is not	Participant describes it as advisory, not a determination by the authority
List what the design will need before submission	Participant uses the approval roadmap rather than guessing

Tasks four and five are the ones that matter, and they test the two biggest risks in this component. If a user reads a finding as a determination, the component has created a liability instead of removing one, however good the rules are. If the roadmap is not usable, the procedural half of the store is dead weight. I report both as they came out, including where the result is unflattering.

Two substantive changes have come out of the sessions so far. The buildable envelope was originally reported only as a number, and participants consistently read it as a limit on floor area rather than on footprint; it is now reported with the footprint wording and the clearances that produced it. The advisory wording was moved from a footer onto the finding itself after the first session, in which a participant read a compliant verdict as approval. One concern remains open: too few participants have been through the protocol for me to claim that the roadmap is understood by somebody who has never submitted an application, and Section 7.5.5 carries that forward.

6.8 Integration results

End to end, the component accepted Component 1 and Component 2 output without manual intervention once the unit and ordering defects of Section 5.3 were fixed. In the group's representative walkthrough a coastal plot and a two-storey residential plan produced a site block, a geometry block, a compliance verdict and then a prediction, a capital cost and a two-specification comparison, with no human step between the pin and the report.

My integration finding is the one in Section 6.3: the compliance constraint changes the floor area Component 4 prices in three designs out of five. A platform that ran the cost engine on the design as drawn would be pricing a building that cannot be approved, and the number it produced would be wrong in the direction that matters, because it would be too low.

6.9 Achievement of objectives

Table 6.7 maps each objective from Section 1.7 to the evidence for it.

Table 6.7: Objectives and the evidence for each

Objective	Evidence	Status
O1 Select the low-rise subset of the regulations	Twenty rules in four families, Table 4.3	Achieved
O2 Formalise the subset as a declarative store	Store edited without code change, Section 5.2.1	Achieved
O3 Implement a three-valued evaluator with citation	100 % of findings clause-cited; zero unsafe passes	Achieved
O4 Derive envelope and roadmap and publish them	Regulatory block, Table 4.4; Sections 6.1.2, 6.6	Achieved
O5 Verify against hand-worked cases and the gazette	Twelve reference cases; 144 of 144 cells agreeing	Achieved
O6 Quantify which rules bind and what the constraint removes	Sections 6.1.4, 6.2, 6.3	Achieved, with two rules unexercised

7 DISCUSSION

7.1 Interpretation of findings

Component 3 does what it was built to do. A coordinate, a floor plan and a zone go in; a clause-cited planning position, a buildable envelope and an approval roadmap come out, in under a millisecond, with the effective date of the rules attached. RQ1 is answered affirmatively: the low-rise subset of these regulations is expressible as data that a generic evaluator can run, and no clause needed hand-written code.

Of everything in Chapter 6 the engineering result is the least interesting, and I want to be explicit about why. A rule engine that evaluates twenty rules correctly is an undergraduate exercise. What the project is for is the three findings that came out of running those rules over a population, and two of them run against what I expected before I started.

First, the rules that bind are not the rules anybody talks about. Density fires in 14.0 percent of cases and sits ninth; clearances, provisions and room dimensions all fire more often. Every public conversation about planning in Sri Lanka is about floor area ratio, and on an ordinary plot the median design has 0.636 of a ratio point to spare while sitting 0.9 metres inside its side clearance.

Second, clearance is what makes a small plot unbuildable, not density. At the statutory minimum extent of 150 square metres a two-storey building can cover a quarter of the land and a three-storey building can cover a twentieth. That is a consequence of banding: crossing 7 metres of height moves the whole building into a stricter row of Form E, and the regulation gives no warning that the step is there.

Third, the compliance constraint is not cosmetic for the platform. It removes 22.4 percent of drawn floor area on average, and leaves 39.3 percent of designs untouched. A cost figure computed before the compliance check is a cost figure for a building that may not be approvable.

7.2 Comparison with literature

Against the model-based systems [4], [6], [7] the comparison is not flattering on capability and is not meant to be. Those systems check hundreds of clauses against a

semantically rich model; mine checks twenty against a dozen scalars. What it does that they do not is run on inputs that exist in this market, and report honestly when they do not.

Against the extraction literature [5], [12], [13] my position is a deliberate step backwards. Those systems reduce the cost of getting rules out of text; I paid that cost manually and bought auditability with it. At twenty rules that is clearly the right trade. At two hundred it would clearly be the wrong one, and the release of annotated corpora such as CODE-ACCORD [8] is what will eventually make the automated route defensible for a jurisdiction like this one.

On the open-world point I could not find a comparable treatment. The systems I read report pass or fail; none of them reports that a check could not be made and names the missing input. That may be because a BIM model rarely lacks a quantity, in which case the three-valued design is a consequence of my inputs rather than an improvement on theirs. I think it is more general than that, because any checker fed by measurement has the same exposure, but I cannot demonstrate it from one component.

7.3 Reflection on the design decisions

Putting the rules in data was right and I would do it again. It cost an afternoon of extra abstraction and it is the only reason a planner could maintain the store.

Naive forward chaining was also right, and Section 6.5 says so by three orders of magnitude. If I had reached for a Rete network because the literature mentions one [10], I would have spent a week on a problem the measurement shows does not exist.

Manual transcription I am less sure about. It worked, and the double transcription caught the errors I made, but it does not scale past this store and it makes the component expensive to retarget at a second jurisdiction. The honest position is that it was right for this project and wrong as a product strategy, which Chapter 8 has to live with.

The rectangular envelope approximation is the decision I would change first. It was a simplification I made in week two to get something working, and it survived to submission because nothing forced it to change. A corner plot with two frontages

gets a figure that is indicative at best, and users in the acceptance sessions treated the envelope as the most concrete number the component produced, which is exactly the wrong combination.

7.4 Implications

For practice the implication is that a planning screen is cheap enough to be free. Nothing in the component needs paid data or specialist hardware, and the evaluation cost is negligible, so there is no technical reason a property portal could not run this on every listing. The barrier is maintaining the store, not running it.

For research the narrower implication is more interesting. Compliance-checking papers report coverage of a code and correctness on demonstration cases. Almost none report the distribution of outcomes over a population, and the distribution is where the surprising content is: it tells you which clauses are doing the work, which have pass bands so wide they are decorative, and which cannot be exercised at all. That analysis costs very little once the rules are encoded, and I would argue it should be standard.

7.5 Limitations and threats to validity

7.5.1 The corpus is constructed, not collected

This is the limitation that bounds everything else. I could not obtain real submissions with recorded municipal outcomes, so the corpus is generated and its sampling is centred near the thresholds by my own choice. Every firing rate in Chapter 6 is therefore a statement about the rules and about my sampling, jointly, and none of them is a statement about how often Sri Lankan designs breach the regulations. The way out is a corpus of real applications with their outcomes, which needs an agreement with a local authority and is the first item in Section 9.4.

7.5.2 One jurisdiction, one snapshot, one subset

The store holds the low-rise subset of one jurisdiction's regulations as published on 8 July 2021. Clauses that could not be decided from quantities the platform can obtain were excluded, which means the component can report a design as free of findings while that design fails something the component never looked at. The interface says so, and the effective date is on every verdict, but a user who reads a clean result as a

clean bill of health has been misled by the product even though the wording is accurate.

7.5.3 Two rules were never exercised, and two rest on interpretations

Minimum plot depth and the qualified person's report never fired across the corpus, so they are encoded and untested. Separately, the storey caps on narrow access and the slope threshold for geotechnical clearance rest on readings I made where the schedule was indexed by something Form C does not publish, or printed in a form that admits more than one interpretation. Both are flagged as provisional in the store and both should be confirmed with a planning officer before anybody relies on them.

7.5.4 The envelope is a rectangle

The buildable envelope assumes a rectangular plot and uniform clearances. Irregular plots, corner plots with two street frontages and plots with an existing building to work around all get a figure that is indicative. Since the acceptance sessions showed this is the number users trust most, the gap between how reliable it is and how reliable it looks is the sharpest usability risk in the component.

7.5.5 Acceptance evidence is thin

Five tasks and a small number of participants support the wording changes in Section 6.7 and little else. In particular I cannot claim that the approval roadmap is intelligible to somebody who has never submitted an application, which is precisely the user the component is for.

7.5.6 Garbage in, confident garbage out

Everything the component says rests on geometry measured by Component 2 and a zone chosen by the user. Section 6.2 bounds the first exposure and shows it is modest away from the limit and decisive near it. The second is not bounded at all: a user who picks the wrong density category gets a confident, clause-cited, entirely wrong entitlement, and the component has no way to detect it.

7.6 Summary of the discussion

What the component establishes is that this regulation is formalisable, that the formalisation is cheap to run and auditable to read, and that running it over a population tells you things about the regulation that reading it does not. What it does

not establish is anything about real submissions, and I would rather be clear about that than let an aggregate figure do work it cannot support.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

Stated without marketing language: the component tells a buyer, before they pay, what they are allowed to build on the land and what the approval will require of them. Table 8.1 sets out the four groups that would pay for that, directly or through the platform.

Table 8.1: Beneficiaries of the compliance component

Group	What they get	Why they would pay
Land buyers	Entitlement and envelope before purchase	The decision with the largest financial consequence is the one taken with the least information
Small practices	A pre-submission screen in seconds	A returned application costs weeks of calendar time [2]
Property portals	A planning view on every listing	Differentiates a listing page at negligible marginal cost
Local authorities	Fewer submissions failing on basic rules	Officer time spent on judgement instead of arithmetic

8.2 Market alternatives and differentiation

Table 2.1 set out the alternatives. The short version is that they are either authoritative and late, or free and unusable. An architect gives a better answer than my component and gives it after you have engaged an architect; the gazette gives the complete answer and gives it in a form nobody can apply to their own plot.

Defensibility is weak on technology and reasonable on currency. Any competent developer could encode twenty rules in a fortnight. What is hard to copy is a store that is actually kept current against the gazette, and that is a commitment rather than a technical asset.

8.3 Business model and feasibility

We settled on freemium business-to-consumer with paid business-to-business tiers. A homeowner checks one plot free; a practice pays a subscription for bulk checks and the pre-submission report; a portal pays for an integration. My component suits

that model well on the cost side, because the data is a document rather than a feed and the evaluation is effectively free.

The binding constraint is not technical. It is that somebody has to read the gazette when it moves and edit the store, and that person has to understand planning law well enough to know what an amendment changes. A platform that stops doing that does not degrade gracefully; it keeps answering, confidently, with rules that have been superseded. That is the one operational commitment the business cannot defer, and it is why NFR5 puts the effective date on the face of every verdict.

8.4 Intellectual property and compliance

The regulations are a public instrument and there is no restriction on encoding them. The store, the evaluator and the corpus generator are the group's own work. Two exposures need naming. A compliance finding that a user relies on, and that turns out to be wrong because the store was stale, is a professional liability question rather than an intellectual property one, and the advisory wording and the effective date are the mitigations. And a commercial deployment should take legal advice on the wording of the disclaimer instead of relying on an undergraduate's reading of it.

8.5 Sustainable Development Goal contribution

The plausible contribution is to Goal 11, through fewer unbuildable purchases and fewer returned applications, and marginally to Goal 9 through a less wasteful approval process. I should add the caveat: this is a plausible contribution and not a measured one. No part of this project measured avoided applications or avoided purchases, and claiming otherwise would be the kind of unsupported impact statement the group agreed to avoid.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Summary of the work

In short, Component 3 encodes a working subset of the Urban Development Authority Planning and Development Regulations 2021 as a declarative rule store, evaluates it with a forward-chaining engine that reports compliant, violated or not assessed with the clause behind every finding, and turns the result into a buildable envelope and an approval roadmap that constrain what the rest of the platform prices.

9.2 Answers to the research questions

Table 9.1 states each answer in one line.

Table 9.1: Research questions and the answers reached

Question	Answer
RQ1 Can the subset be expressed declaratively?	Yes. Twenty rules and 144 schedule cells run on a generic evaluator with no clause-specific code; a regulation change is a data edit.
RQ2 Which rules bind?	Not the ones the regulations lead with. Rainwater storage, side space, parking, rear space and room dimensions all fire more often than floor area ratio, which sits ninth at 14.0 percent.
RQ3 How sensitive is the density verdict?	Robust away from the limit and decisive near it: 2.7 percent of verdicts flip at 5 percent measurement error, 8.1 percent at 20 percent. The component reports the margin for that reason.
RQ4 What does compliance remove?	22.4 percent of drawn floor area on average, nothing at all for 39.3 percent of designs, and more than 65 percent for the worst tenth.

9.3 Contribution restated

The engineering contribution is a maintainable encoding of a South Asian planning regime, which as far as I can find does not otherwise exist in the literature.

What I did not expect, and what I think is worth more than the engineering, is the population view. Encoding the rules took a few weeks; running them over a thousand cases took an afternoon and showed that the clause everybody argues about has the widest pass band of any rule in the store, while the clause that makes small plots undevelopable is a clearance figure in a schedule nobody reads.

9.4 Future work

- **Build the corpus of real applications.** Submissions with their recorded outcomes, under an agreement with a local authority. Without it every firing rate in Chapter 6 remains a statement about my sampling.
- **Replace the rectangular envelope with real plot geometry**, taking the boundary polygon Component 1 already holds instead of a frontage and a depth. This is the clearest unforced weakness in the component.
- **Confirm the two provisional readings** with a planning officer, and exercise the two rules the corpus never triggered.
- **Semi-automate the store** once it passes a few hundred rules, in the direction the extraction literature indicates [8], [13], with the output reviewed by hand before it enters the store.
- **Add the local-authority layer.** The store encodes the national regulations; municipal by-laws and fee schedules sit on top of them and vary by authority.
- **Report an uncertainty band on the density verdict**, derived from the measurement error Component 2 declares, instead of a margin the user has to interpret.

9.5 Closing remark

A regulation that is public but unusable is not meaningfully public. That sentence is the whole motivation for this component, and the work of the last several months has been an argument that the distance between the two is a few weeks of careful transcription and a few hundred lines of evaluator.

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GLOSSARY

Buildable envelope – The footprint of a plot that remains once the statutory front, rear and side spaces have been deducted.

Density category – The low, medium or high classification of a zone, which selects the column group of Schedule 6 Form C.

Derived fact – A quantity the engine computes from asserted facts, such as the permissible floor area ratio for a plot.

Effective date – The date on which the regulation behind a rule came into force; carried on every finding the component produces.

Floor area ratio – Gross floor area, excluding parking, divided by the extent of the plot.

Forward chaining – Deriving new facts from asserted ones until no further fact can be produced, then evaluating rules against the closure.

Not assessed – The verdict returned when a rule's inputs are incomplete; distinct from compliant and from violated.

Preliminary planning clearance – A clearance required before a development above stated thresholds may be submitted for approval.

Rule store – The declarative document holding the encoded rules and the numeric schedules they refer to.

Site coverage – The footprint of the building divided by the extent of the plot.

Statutory space – The open space a regulation requires between a building and the boundary or the street.

Street line – The declared line from which the front space of a building is measured; where none is declared, the edge of the road is used.

APPENDICES

Appendix A Supporting material

This appendix carries the material that supports Chapters 3 to 6 without belonging in the main text.

A.1 The encoded rule set

Table A.1 lists every rule in the store with the clause it encodes and the severity it carries. Severity is what lets the interface separate a density breach, which may stop the project, from an unmet rainwater provision, which is a line item in the drawing set.

Table A.1: The twenty encoded rules

ID	Family	Requirement	Severity
D1	Density	Floor area ratio within the permissible value for the plot extent, road width and density category	Major
D2	Density	Site coverage within the permissible fraction for the height band and use	Major
D3	Density	Storeys within the cap imposed by the width of the access	Major
M1	Dimensional	Plot extent not less than 150 m²	Major
M2	Dimensional	Frontage to a street not less than 6.0 m	Major
M3	Dimensional	Plot depth not less than 12.0 m	Minor
M4	Dimensional	Habitable rooms meeting the minimum area and least width of Schedule 7	Minor
M5	Dimensional	Clear room height not less than the minimum for the use	Minor
M6	Dimensional	Boundary wall not exceeding 2.0 m	Minor
S1	Spatial	Rear space not less than the minimum for the height band	Major
S2	Spatial	Side space not less than the minimum, and never less than 1.0 m	Major
S3	Spatial	Where no street line is declared, 1.0 m from the edge of the road	Major
S4	Spatial	Light well not smaller than the minimum for the height band	Minor
S5	Spatial	Off-street parking provided per Schedule 10	Minor
P1	Procedural	Preliminary planning clearance above 1 000 m² or 15 m	Obligation
P2	Procedural	Fire clearance above 15 m or five dwelling units	Obligation
P3	Procedural	Green building certificate above 1 000 m²	Obligation
P4	Procedural	Geotechnical clearance where the land slope exceeds the threshold	Obligation

ID	Family	Requirement	Severity
P5	Procedural	Qualified person's report for a site above 0.5 hectares	Obligation
P6	Procedural	Rainwater storage at 1.5 m³ per 100 m² of roof area	Obligation

A.2 Verification against the gazette

Schedule 6 Form C was transcribed twice, once from the gazette text and once from the consolidated schedule tables, and the two transcriptions were compared cell by cell. All 144 cells agreed. Twelve reference cases were then worked by hand from the gazette, covering each rule family and placing at least one case on each side of every band boundary, and the engine agreed with all twelve. Those cases are committed as data and asserted in the test suite, so a future edit to the store that breaks one of them fails the build.

A.3 Corpus generation procedure

The corpus was generated with a fixed seed from the distributions in Table 3.2. Plot extent is drawn log-normally and clipped, frontage is drawn relative to the square root of the extent, and depth follows from the two so that the three remain mutually consistent. Every quantity that a rule tests is centred close to the statutory threshold, with a spread chosen so that roughly one case in five to one in three falls on the violating side. That centring is a design choice and it is the reason the corpus exercises every rule; it is also the reason the firing rates say nothing about real submissions.

A.4 Interpretation decisions recorded in the store

Two rules rest on a reading rather than a transcription, and both are flagged in the store so that a reviewer sees them.

The first concerns the storey caps on narrow access. Form C applies where no zoning-factor development plan exists, but the caps in Form B are indexed by the zoning factor that such a plan would publish. The store maps the three Form C density categories onto the factor bands, so that a controlled zone on a 3.0 metre access is capped at one storey, a residential zone at two and a mixed zone at three, with the caps rising by one on a 4.5 metre access. The second concerns the ground slope above which geotechnical clearance attaches, which the consolidated schedule

prints in a form that admits more than one reading; the store takes 11 degrees. Neither reading should be relied on without confirmation from a planning officer, and Section 7.5.3 says so in the main text.

A.5 Acceptance protocol

Participants were given the five tasks in Table 6.6 in the order shown, each on a plot or project they knew personally so that their own knowledge of the site supplied the reference rather than anything the system produced. An observer who did not assist recorded task completion, the time taken, points of confusion and verbatim comments. The fourth task was treated as the critical one, because the risk with a compliance finding is that a user reads it as a determination; a participant who came away believing the platform had approved their design would represent a failure of the wording regardless of the quality of the rules.

Appendix B Academic integrity and AI use disclosure

In accordance with the module policy the author discloses that generative artificial intelligence assistance was used during the preparation of this thesis and its associated artefact. That assistance was confined to three kinds of support: help in diagnosing defects in the evaluator and the analysis code described in Sections 5.2 and 6.1, language editing and proofreading of draft chapters, and an initial triage of candidate literature for Chapter 2. Every suggested change to the code was read, adapted and tested by the author before being adopted; every technical claim was confirmed independently and every altered passage rewritten in the author's own words; and every cited source was retrieved and read in full by the author. No part of the rule store was produced by a generative tool: every threshold in it was transcribed by hand from the gazette and checked twice, for the reasons given in Section 4.3.4.

No generated text was submitted without review. All results, figures and tables in Chapter 6 were produced by committed scripts run by the author against the committed rule store and corpus with a fixed random seed, and the author can reproduce and defend every number reported.

B.1 Statement of original contribution

The regulation review, the rule selection and transcription, the rule store, the forward-chaining evaluator, the buildable envelope and approval roadmap, the corpus design, the evaluation and the written analysis in this thesis are the author's own work. Contributions from the other three group members are identified where they appear: the plot geometry and terrain figures come from Component 1, the design geometry from Component 2, and the regulatory contract was agreed jointly with the Component 4 owner, who consumes the permitted floor area this component publishes.